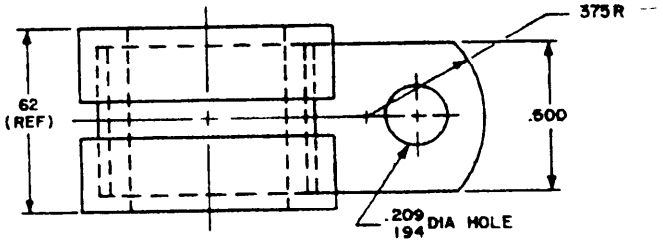
MIL-C-85052/1B
30 MAY 1984
SUPERSEDING
MIL-C-85052/1A
15 SEPTEMBER 1980

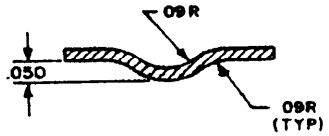
MILITARY SPECIFICATION SHEET

CLAMP, LOOP, TUBE-
17-7 PH CRES, 275° F, FUEL AND PETROLEUM BASED
HYDRAULIC FLUID RESISTANT

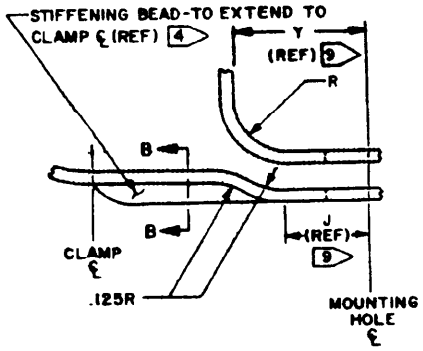
THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE

THE COMPLETE REQUIREMENTS FOR ACQUIRING THE CLAMPS DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE LATEST ISSUE IN EFFECT OF SPECIFICATION MIL-C-85052

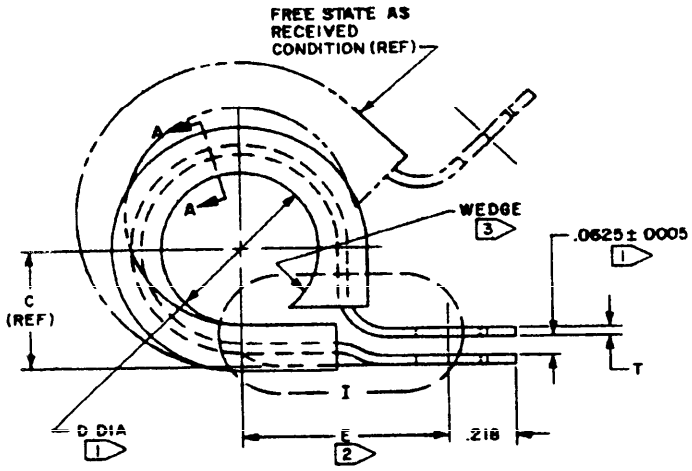




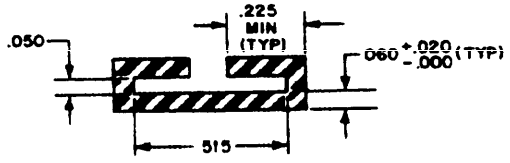
SECTION B-B



DETAIL I
(CUSHION REMOVED)



TOLERANCES .XXX ± .010, .XX ± .03 UNLESS OTHERWISE SPECIFIED



SECTION A-A
(CUSHION ONLY)

Ⓑ DENOTES CHANGES

FSC 5340

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TABLE I

DASH NUMBER	C (REF)	D DIA (1)	(B) E ±.032 (2)	J (REF)	R ±.010	T	Y (REF)
2	192	125	468	--			
3	224	188	499	--			
4	255	250	530			.020	
5	286	312	561		090	± 002	325
6	318	375	592	235			
7	349	438	624				
8	380	500	655				
9	423	562	741				
10	454	625	772			.032	
11	486	688	804	256	125	± 002	363
12	517	750	835				
13	548	812	866				

DASH NUMBER	C (REF)	D DIA (1)	(B) E ±.032 (2)	J (REF)	R ± 010	T	Y (REF)
14	580	.875	898				
15	611	.938	929	256		.032	368
16	642	1 000	960			± 002	
17	681	1 062	1 001				
18	712	1 125	1 032				
19	744	1 188	1 064		125		
20	775	1 250	1 095			.040	
21	806	1 312	1 126	262		± 003	370
22	838	1 375	1 158				
23	869	1 438	1 189				
24	900	1 500	1 220				

NOTES

(1) DIAMETER "D" IS THE NOMINAL DIAMETER FOR WHICH A CLAMP SIZE IS INTENDED FOR USE. DIAMETER "D" SHALL BE VERIFIED BY THE "DIAMETRAL RETENTION" TEST OF THE GENERAL SPECIFICATION.

(2) DIMENSION "E" SHALL BE MEASURED WITH THE CLAMP INSTALLED ON A MANDREL OF "D" DIAMETER ± 001 AND AN .0625 ± 0005 SPACER BETWEEN THE UPPER AND LOWER FOOT AS SHOWN.

(B) (3) WEDGE SHALL BE REQUIRED ON SIZE -8 AND LARGER. THE WEDGE SHALL BE MOLDED AS AN INTEGRAL PART OF THE CUSHION AND CONTOURED TO FIT "D" DIAMETER.

(B) (4) STIFFENING BEAD SHALL BE REQUIRED ON SIZE -4 AND LARGER. ALL RADII ON THE STIFFENING BEAD SHALL BE SMOOTH AND BLENDED. NO SHARP TOOL MARKS ARE ALLOWED.

(B) 5. UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.

6. METAL BANDS SHALL HAVE ALL BURRS, SHARP EDGES AND SCALE REMOVED.

(B) (7) WEDGE SHALL TOUCH CUSHION ON LOWER FOOT WITH CLAMP INSTALLED ON MANDREL WITHOUT SPACER.

(B) (8) IF MIL-C-85052/4 CLAMP SUPPORTS ARE USED, THE MAXIMUM TUBE OVERHANG SHALL BE 1/2 TUBE O.D. THIS WASHER CAN BE DETRIMENTAL IF THE CLAMP IS NOT FULLY SUPPORTED.

(9) REFERENCE DIMENSIONS "Y" AND "J" ARE PROVIDED TO GAIN MAXIMUM SUPPORT FOR THE LOWER FOOT BY CLOSELY FITTING THE UPPER FOOT BEND RADIUS TO THE STIFFENING BEAD BLEND RADIUS WHEN CLOSED AGAINST EACH OTHER.

(10) UNLESS OTHERWISE SPECIFIED, PROPERTY MEASUREMENTS SHALL BE PERFORMED IMMEDIATELY AFTER REMOVAL FROM THE IMMERSION BATH.

REQUIREMENTS

MARKING

BAND - THE COMPLETE STANDARD PART NUMBER AND MANUFACTURER'S NAME, TRADEMARK OR FSCM NUMBER SHALL BE IMPRESSION STAMPED ON THE BAND IN AN AREA NOT COVERED BY THE CUSHION. ALL MARKING SHALL BE ABOVE TUBE THEORETICAL CENTERLINE.

- CLAMP BANDS OF -2, -3, -4 SIZES MAY BE MARKED WITH THE MANUFACTURERS INDENTIFICATION, THE SIZE, AND THE SPECIFICATION SHEET NUMBER DUE TO SPACE LIMITATIONS.

CUSHION - NONE

MATERIALS

METAL BAND - 17-7PH, CORROSION RESISTANT STEEL PER MIL-S-25043, ANNEALED, STRESS RELIEVE AND HEAT TREAT TO TH1100 CONDITION PER MIL-H-6875, AFTER FORMING.

CUSHION - NITRILE BUTADIENE RUBBER, 65-75 DUROMETER, COLOR - YELLOW, PER THE GENERAL SPECIFICATION WITH ADDITIONAL REQUIREMENTS SHOWN ON THIS SPECIFICATION SHEET.

FINISH

METAL BAND - PASSIVATE PER QQ-P-35

CUSHION - NONE

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T NUMBER

PART NUMBERS SHALL CONSIST OF THE FOLLOWING (IN SEQUENCE)

1 THE LETTER "M"
2 THE GENERAL SPECIFICATION NUMBER
3 A SLASH AND SLASH NUMBER OF THIS SPECIFICATION SHEET
4 A DASH, AND THE APPROPRIATE SIZE DASH NUMBERS FROM TABLE I

EXAMPLE M85052/1-8

CUSHION REQUIREMENTS

1 PHYSICAL PROPERTIES - AS SPECIFIED IN TABLE II UNLESS OTHERWISE SPECIFIED, RESULTS ARE AN AVERAGE OF 5 SPECIMENS AND TOLERANCE ON TEMPERATURE = ± 5°F

TABLE II (B)

TEST	TEST METHOD	REQUIRED ORIGINAL PROPERTIES	ALLOWABLE CHANGE FROM ACTUAL ORIGINAL PROPERTIES AFTER		
			FUEL IMMERSION MIL-T5624 JP-4 168 HRS AT +100°F (10)	HYDRAULIC FLUID IMMERSION MIL-H-5606 70 HRS AT +158°F (10)	HEAT AGING 70 HRS AT 275°F
HARDNESS, DUROMETER "A"	ASTM D2240	65-75	-20 MAX	-15 MAX	+10 MAX
TENSILE STRENGTH - PSI	ASTM D412	2000 MIN	-30% MAX	-10% MAX	-40% MAX
ELONGATION - %	ASTM D412	500 MIN	-10% MAX	-10% MAX	-75% MAX
TEAR STRENGTH - PPI	ASTM D624 DIE "B"	300 MIN	-35% MAX	-10% MAX	-40% MAX
SPECIFIC GRAVITY	------	AS MEASURED	------	------	±2%
VOLUME CHANGE - %	ASTM D471-75				
1 AFTER CONDITIONING		------	+15% MAX	+10% MAX	-10% MAX
2 AFTER 24 HRS AIR DRY		------	+10% MAX	+10% MAX	------

(B) 2 OZONE RESISTANCE - REQUIRED PER MIL-C-85052.

(B) 3 COMPRESSION SET - SEE MIL-C-85052

- AIR AGE AT 212 F.
- NOT TO EXCEED 55% AVERAGE OF 3 SPECIMENS

(B) 4. FLAMMABILITY - SEE MIL-C-85052

- SPECIMENS: -16 SIZE CLAMP ASSEMBLIES UNBENT UNTIL FLAT
- VERTICAL BURN TEST

ED USE

THESE CLAMPS ARE INTENDED FOR USE AS FOLLOWS

TEMPERATURE RANGE

-65°F TO +275°F (FOR HIGHER TEMPERATURE RATED CLAMPS, SEE MIL-C-85052/3 CLAMPS)

SYSTEMS

1 ALL AROMATIC FUEL SYSTEM APPLICATIONS REQUIRING A HIGH DEGREE OF CUSHION EXPOSURE TO SYSTEM FLUID

2. ALL FLUID AND ELECTRICAL SYSTEMS ON AIRCRAFT AND AEROSPACE VEHICLES USING PETROLEUM BASED HYDRAULIC SYSTEMS

VIBRATION RATING

DEPENDING ON MOUNTING METHODS (SEE RECOMMENDED MOUNTING DATA)

Custodians:
NAVY-AS
AIR FORCE-99
ARMY-AV

Preparing activity
NAVY-AS

PROJECT NO 5340-1572

Reviewer activities
AIR FORCE - 11
DLA - IS

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RECOMMENDED MOUNTING METHODS

MOUNTING METHODS AFFECT THE VIBRATION CAPABILITIES OF THESE CLAMPS TYPICAL MOUNTING METHODS AND VIBRATION RATINGS ARE SHOWN IN FIGURES I, II, AND III THE METHOD SHOWN IN FIGURE II IS RECOMMENDED WHENEVER POSSIBLE THE METHOD SHOWN IN FIGURE III SHOULD ONLY BE USED IN VERY HIGH VIBRATION AREAS OR AS NECESSARY

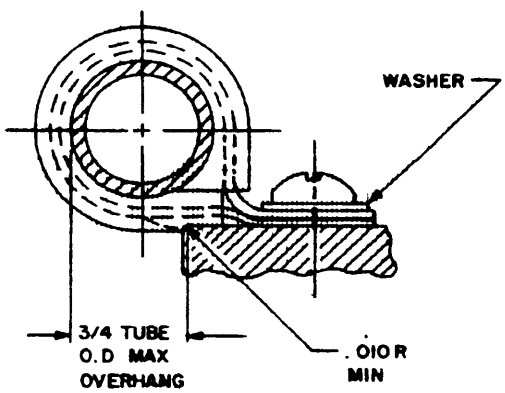


FIGURE I
LOW TO MEDIUM VIBRATION AREAS

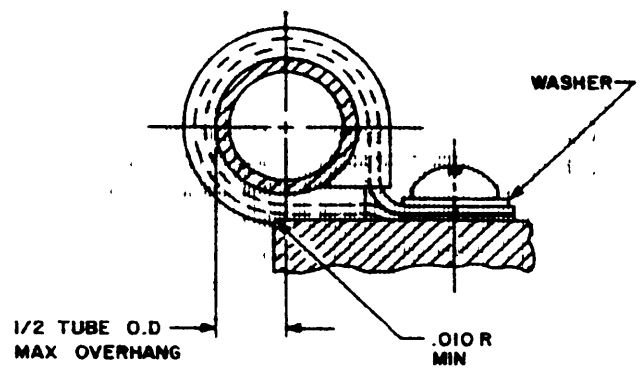


FIGURE II
LOW TO HIGH VIBRATION AREAS

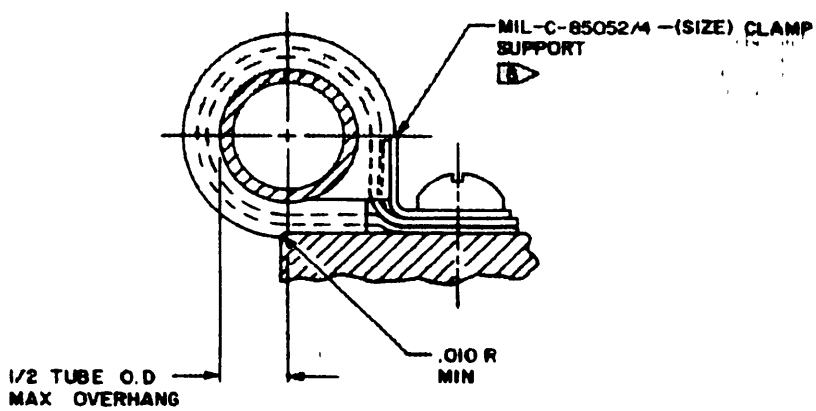


FIGURE III
VERY HIGH VIBRATION AREAS

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INCH–POUND

NOTICE
OF VALIDATION

MIL–C–85052/1B
NOTICE 1
26 September 1991

MILITARY SPECIFICATION

CLAMP, LOOP, TUBE 17–7 PH CRES, 275 DEG. F, FUEL AND PETROLEUM BASED HYDRAULIC FLUID RESISTANT

MIL–C–85052/1B, dated 30 May 1984, has been reviewed and determined to be valid for use in acquisition.

Custodians:
Army – AV
Navy – AS
Air Force – 99

Preparing activity:
Navy – AS

AMSC N/A

FSC 5340

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